

# MIDE: Multimodal Dialogue Emotion Recognition via Mutual Information Enhancement and Dynamic Modality Selection

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## Abstract

Multimodal dialogue emotion recognition seeks to integrate information from text, audio, and video to accurately determine the emotional state of each utterance. However, prevailing approaches often depend on fixed fusion strategies, failing to account for quality variations across modalities. Consequently, noise from less informative modalities can degrade overall performance. Moreover, most models treat utterances as independent units for static classification, overlooking the dynamic evolution of emotions throughout a dialogue. To overcome these limitations, we propose MIDE, a novel approach that leverages mtual information enhancement and dynamic modality selection for multimodal emotion recognition. Our approach dynamically selects high-quality modality pairs for fusion, minimizing interference from noisy or redundant sources. Specifically, it employs mutual information maximization to achieve cross-modal semantic alignment and incorporates an autonomous modality selection mechanism to assess inter-modal compatibility. Furthermore, an emotion transition prediction module, implemented with a Gated Recurrent Unit (GRU), captures temporal emotional dependencies, enabling joint optimization of static emotion classification and dynamic emotion trajectory prediction. Extensive experiments on the IEMOCAP and MELD datasets demonstrate that MIDE significantly surpasses existing models in accuracy and robustness, highlighting its strength in adaptive fusion for complex dialogue scenarios.

## CCS Concepts

• Computing methodologies → Artificial intelligence; • Information systems → Multimedia information systems.

## Keywords

Sentiment Analysis and Opinion Mining; Dialogue Emotion Recognition; Multimodal; Mutual Information

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## 1 Introduction

With the rapid advancement of artificial intelligence, multimodal dialogue emotion recognition (MDER) [3, 13] has emerged as a critical research area at the intersection of natural language processing, computer vision, and audio analysis. The core objective of MDER is to integrate cross-modal information for accurate identification of emotional states (e.g., happiness, anger) in dialogue utterances, overcoming the limitations of unimodal approaches that overlook inter-modal emotional correlations. Beyond advancing fundamental AI research in multimodal fusion, MDER holds significant practical value in applications such as empathetic human-computer interaction, intelligent customer service, and mental health monitoring, attracting growing interest from both academia and industry [24, 26].

Existing studies generally confirm that multimodal fusion (e.g., text, audio, and video) can compensate for the shortcomings of unimodal methods, thereby improving recognition performance. However, multimodal dialogue emotion recognition still faces several core challenges. **First, modality fusion strategies remain rigid:** most methods adopt fixed weights or manually preset combinations to uniformly integrate multimodal information, failing to account for variations in modality quality and relevance. This inflexibility limits their adaptability to real-world scenarios where modality quality fluctuates dynamically (e.g., due to noise, occlusion, or semantic absence). Such “forced fusion” often introduces redundant or even detrimental features, degrading overall recognition performance. **Second, cross-modal semantic gaps persist:** text, audio, and video modalities differ substantially in feature distributions and forms of semantic expression. Most existing fusion methods rely on superficial feature concatenation, lacking effective mechanisms to quantify inter-modal semantic associations, which hinders deep semantic alignment. **Third, temporal characteristics of dialogues are overlooked:** emotions in dialogue exhibit strong contextual dependencies. Yet, most current approaches process utterances independently, disrupting the dynamic transitions of emotional states and leading to incoherent predictions in longer dialogues.

To address these challenges, this paper proposes MIDE, a novel multimodal dialogue emotion recognition method that integrates mutual information enhancement with dynamic modality selection. Our approach introduces innovations in three key aspects: First, it employs a **mutual information maximization** strategy to quantify semantic relevance between modality pairs, serving as

the foundation for dynamic modality selection. This enables the model to adaptively prioritize high-value modalities during fusion. Second, it incorporates **mutual information-guided attention** within a Transformer-based cross-modal fusion framework, balancing self-attention within individual modalities and cross-attention between modalities to enhance semantic alignment. Third, it designs an **emotion transition prediction module** based on Gated Recurrent Units (GRUs) to explicitly model the temporal evolution of emotions, enabling joint optimization of static emotion classification and dynamic transition prediction. Extensive experiments on the IEMOCAP and MELD datasets demonstrate that our method significantly outperforms state-of-the-art baselines in recognition accuracy, robustness, and temporal modeling capability, underscoring its research value and application potential in complex dialogue scenarios. The main contributions of this work can be summarized as follows:

- We propose a dynamic modality selection mechanism based on mutual information enhancement, which overcomes the limitations of fixed fusion strategies by enabling the model to adaptively focus on highly correlated modality pairs.
- We design a cross-modal Transformer fusion framework that incorporates mutual information weights into cross-attention mechanisms, significantly improving cross-modal semantic alignment.
- We introduce an emotion transition prediction module that facilitates joint optimization of static emotion recognition and dynamic emotion trajectory prediction, effectively capturing temporal emotional dependencies in dialogues.
- Extensive experiments demonstrate that our model not only achieves state-of-the-art performance but also exhibits superior generalization and robustness across datasets and emotion categories.

## 2 Related Work

Emotional Recognition in Conversations (ERC) research can be divided into two categories: unimodal and multimodal. The former focuses on modeling context and speakers within a single modality, while the latter breaks through the limitations of unimodal methods by fusing multimodal information.

### 2.1 Unimodal Dialogue Emotion Recognition

Unimodal ERC primarily employs two technical frameworks: Recurrent Neural Networks (RNs) and Graph Convolutional Networks (GCNs). RNN-based methods pioneered effective context modeling. For example, DialogueRNN [17] introduced three specialized GRU variants—global, speaker-specific, and emotion-specific—enabling simultaneous modeling of dialogue context, speaker states, and emotional dynamics. DialogueCRN [7] incorporated cognitive principles, using bidirectional LSTM to capture scenario and speaker context, building global memory, and iteratively integrating emotion cues via multi-turn reasoning, advancing ERC from context understanding to cognitive reasoning. Subsequent work focused on semantic enhancement and long-range dependencies: SCCL [29] enriched context with external knowledge, EmoTransKG [33] used fine-grained attention for emotion-relevant cues, CoE [22] modeled

long-range dependencies with sparse attention, and DVL-CER [30] simulated interactions through speaker-listener role modeling.

In parallel, GCN-based methods have enhanced interaction modeling through graph-structured representations. DialogueGCN [4] and DAG-ERC [21] represented utterances as graph nodes, defining edges based on speaker identity and temporal relationships, then aggregated inter-utterance and speaker interaction contexts via graph convolutional operations. BLC [27] constructed heterogeneous graphs incorporating “utterance-speaker-emotion event” relationships and modeled “belief-desire” interactions through graph convolutional message passing to improve fine-grained recognition accuracy. LECM [14] derived graph structures directly from acoustic representations and optimized audio-only modeling through multi-stage learning.

### 2.2 Multimodal Dialogue Emotion Recognition

Multimodal ERC centers on the principle of “modality complementarity” and has evolved through three distinct developmental phases: basic fusion, scenario adaptation, and knowledge enhancement. In the basic fusion stage, SDT [16] employed intra-modality and inter-modality Transformer architectures to capture interactions, enhancing representations through hierarchical gating and self-distillation. JOYFUL [11] combined Transformer networks with graph contrastive learning to mitigate GNN over-smoothing. CMCF-SRNet [32] refined semantic representations using cross-modal constrained Transformers and semantic graphs, while MPLMM [6] enabled fine-grained, element-wise interactions across text, audio, and video modalities. The scenario adaptation stage introduced more flexible fusion strategies. Frame-SCN [23] employed adaptive weighted fusion to accommodate specific scenarios. PCDS [20] addressed speaker-unknown scenarios by combining modal query attention with clustering techniques. HiMul-LGG [2] filtered features based on modal relevance and optimized cross-modal alignment, while TopicDiff [15] overcame multilingual barriers through shared subspace learning. The current knowledge enhancement stage has focused on incorporating external knowledge and sophisticated learning paradigms. UniMSE [8] established a knowledge-sharing framework with cross-modal contrastive learning for representation alignment. MultiDAG+CL [18] constructed multimodal Directed Acyclic Graphs (DAGs) integrated with curriculum learning. MAGTKD [12] enhanced modalities through prompt learning and knowledge distillation, aggregating information via anchor-gated Transformers. ECERC [31] adopted an “evidence-cause” perspective, implementing multimodal causal reasoning through evidence gating and attention mechanisms.

Compared to previous multimodal ERC approaches, methods in both the basic fusion (e.g., SDT, JOYFUL) and scenario adaptation (e.g., Frame-SCN, PCDS) stages typically employ fixed-weight fusion strategies, making them susceptible to modality quality fluctuations (such as noise in low-quality videos) or addressing cross-modal semantic gaps through limited module/loss designs. In contrast, our approach introduces mutual information as a core innovation, with a dynamic modality selection module that evaluates inter-modal relevance in real-time and adaptively assigns fusion weights, effectively addressing the vulnerability of fixed-weight schemes to low-quality modality interference.

### 3 Methodology

#### 3.1 Problem Definition

The fundamental objective of multimodal dialogue emotion recognition is to achieve deep fusion of textual, acoustic, and visual information, overcoming the limitations of unimodal approaches to accurately identify the emotional state expressed by each utterance in a conversation. Formally, a dialogue comprising  $N$  utterances is represented as  $U = (u_1^m, s_{u_1}), (u_2^m, s_{u_2}), \dots, (u_N^m, s_{u_N})$ , where  $s_{u_i}$  denotes the speaker of utterance  $u_i$  and  $m \in \{t, a, v\}$  corresponds to textual, acoustic, and visual modalities respectively. The core task is to predict the emotion label  $y_i$  (e.g., “happiness”, “anger”) for each utterance  $u_i$  based on multimodal information and speaker context.

#### 3.2 Model Overview

The proposed MIDE model comprises six carefully-designed components that form a complete pipeline from multimodal feature processing to emotion prediction. As shown in Figure 1, these components work together to jointly optimize static utterance-level classification and dynamic emotion transition modeling. Each component will be systematically described in the following six subsections.

#### 3.3 Feature Extraction

To accurately capture emotional representations across modalities, we employ specialized models for feature extraction: textual features  $X^{(t)} \in \mathbb{R}^{B \times N \times D_t}$  are obtained using RoBERTa, acoustic features  $X^{(a)} \in \mathbb{R}^{B \times N \times D_a}$  using openSMILE, and visual features  $X^{(v)} \in \mathbb{R}^{B \times N \times D_v}$  using DenseNet (where  $B$  denotes batch size,  $N$  the number of utterances, and  $D_t, D_a, D_v$  the original feature dimensions per modality). To address dimensional heterogeneity that may lead to imbalanced feature weighting during fusion, we project all modalities to a unified dimension  $D$  using 1D convolutional layers:

$$X^m = \text{Conv1D}(X^{m^\top})^\top \in \mathbb{R}^{B \times N \times D}, m \in \{t, a, v\} \quad (1)$$

This ensures dimensional consistency for cross-modal operations while preserving core emotional information.

#### 3.4 Dynamic Modality Selection

Conventional multimodal fusion methods typically employ fixed-weight strategies, overlooking the crucial aspect of varying inter-modal correlations. In dialogue emotion recognition, for instance, the expression “I am really angry” typically exhibits stronger correlation between the textual token “angry” and acoustic features (e.g., angry tone, volume) than with potentially ambiguous visual cues (e.g., poor camera angle).

To overcome this limitation, we introduce a Mutual Information Estimator (**MI\_Estimator**) to quantify cross-modal correlations, thereby alleviating fusion inefficiencies caused by multimodal heterogeneity. When two modalities carry complementary emotional cues, their mutual information increases, justifying the assignment of higher fusion weights. We further design a **ModalMatcher**, which dynamically computes the average mutual information for each modality pair and employs these values as fusion weights.

This ensures that strongly correlated modalities contribute more substantially to the final representation.

**Mutual information estimation of modality pairs:** The proposed **MI\_Estimator** computes mutual information scores between modality pairs, precisely quantifying their semantic correlation. Specifically, for modalities  $X$  and  $Y$  (post-convolutional features),

$$MI_{xy} = MI(X, Y) = \text{MLP}(\text{cat}(X_{\oplus 1}, Y_{\oplus 1})) \in \mathbb{R}^{B \times N \times N} \quad (2)$$

where  $\text{MLP}$  denotes a multi-layer perceptron, and  $\oplus_k 1$  indicates dimension insertion and repetition along the  $k$ -th dimension. To ensure effective modality selection, we introduce a mutual information loss function,

$$\mathcal{L}_{MI} = -\frac{1}{N} \sum \log \left( \frac{\exp(MI(x_i, y_k)/\tau)}{\sum_j \exp(MI(x_i, y_j)/\tau)} \right) \quad (3)$$

where  $x_i$  and  $y_j$  represent convolutional features, and  $\tau$  is a temperature parameter controlling distribution smoothness.

**Dynamic Selection Weight Generation:** The **ModalMatcher** autonomously evaluates three modality pairs and prioritizes fusion of the highest mutual information pair. Specifically, we compute average mutual information scores  $\bar{MI}_{ta}$ ,  $\bar{MI}_{tv}$ , and  $\bar{MI}_{av}$  by,

$$\bar{MI}_{xy} = \bar{MI}(X, Y) = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N MI(X, Y) \in \mathbb{R}^B \quad (4)$$

The weight optimization process involves threshold-based screening. First, the stacked feature vectors are passed through an  $\text{MLP}_w$  and a Softmax layer to generate initial selection weights that sum to 1:

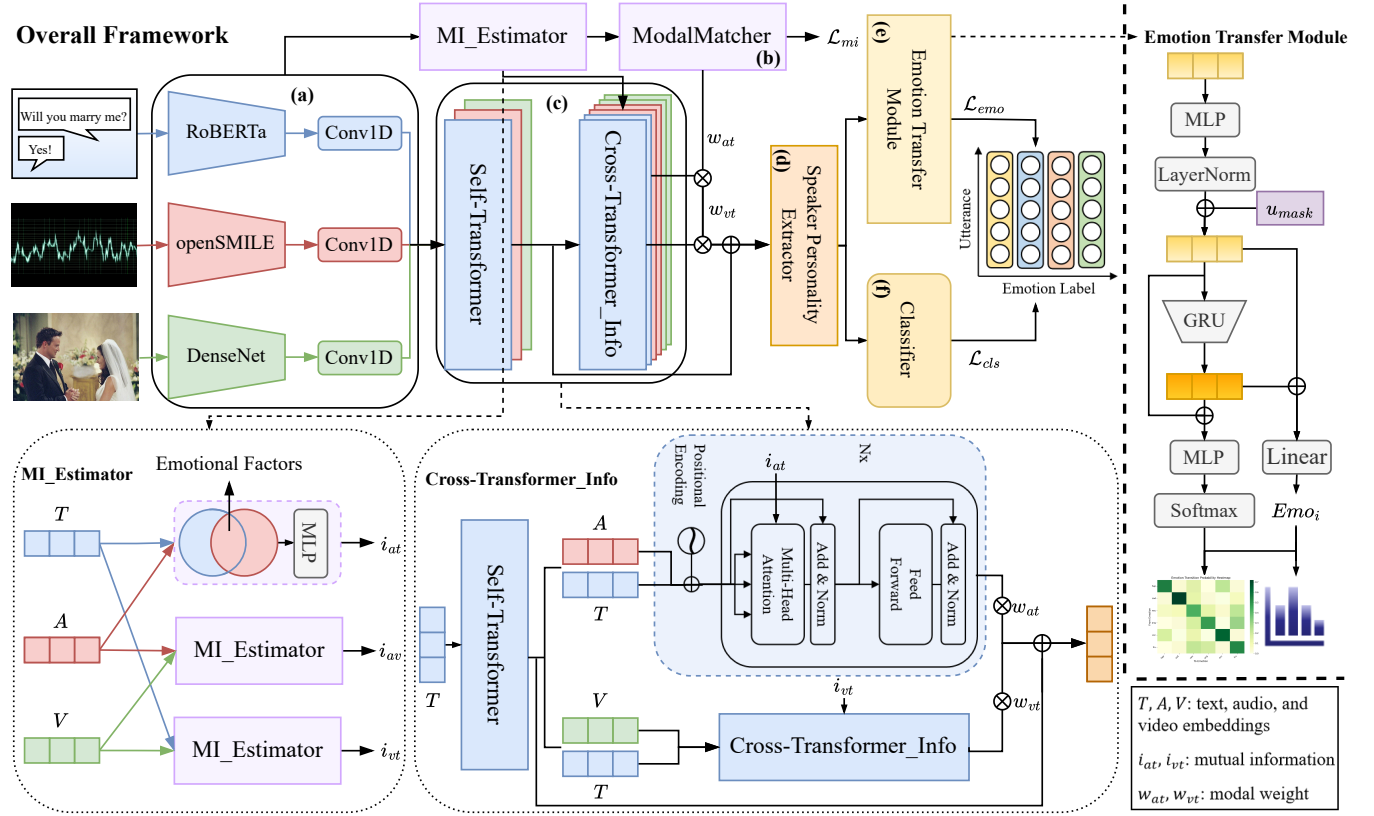
$$[w_{ta}, w_{tv}, w_{av}] = \text{Softmax}(\text{MLP}_w(\text{stack}(\bar{MI}_{ta}, \bar{MI}_{tv}, \bar{MI}_{av}))) \quad (5)$$

A threshold mechanism then refines the weight allocation: weights below  $\frac{1}{3(1+\chi)}$  are set to zero, and the remaining weights are renormalized to sum to 1. If only two modality pairs remain and one weight still falls below  $\frac{1}{2(1+\chi)}$  after renormalization, it is also set to zero. The final output is a set of dynamic selection weights that emphasize high-matching modality pairs. The resulting weight vector  $[w_{ta}, w_{tv}, w_{av}]$  reflects the fusion priorities of the Text-Audio, Text-Visual, and Audio-Visual pairs, respectively. This approach directs the model’s focus to modal regions with high mutual information, significantly improving cross-modal feature interaction. For instance, when text and audio are highly correlated, the Text-Audio weight increases substantially, enabling the fused features to represent emotional content more accurately.

#### 3.5 Mutual Information Enhancement

Based on the Transformer architecture, we design a “self-attention enhancement + weighted cross-attention fusion” mechanism that incorporates modality selection weights to enhance the contribution of highly correlated modality pairs in emotion recognition. The framework consists of Self-Transformer and Cross-Transformer\_Info modes.

Before features are fed into the Transformer encoder, we enhance them to better capture sequential positions and speaker-specific emotional styles. Specifically, positional encoding (PE) and speaker



**Figure 1: The architecture of the MIDE model, composed of six core components: (a) Feature Extraction, (b) Dynamic Modality Selection, (c) Mutual Information Enhancement, (d) Speaker Personality Extraction, (e) Emotion Transition, and (f) Classification. Dashed arrows within each module illustrate key subprocesses.**

embedding ( $s_{emb}$ ) are added to the original feature  $X$ :

$$X' = X + PE + s_{emb} \in \mathbb{R}^{B \times N \times D}; \quad (6)$$

where

$$s_{emb} = \text{Embedding}(s_{u_i});$$

$$PE_{t,2k} = \sin\left(\frac{t}{10000^{\frac{2k}{D}}}\right); PE_{t,2k+1} = \cos\left(\frac{t}{10000^{\frac{2k}{D}}}\right) \quad (7)$$

**Self-Transformer Mode (e.g., Text-Text):** In this case, the input modality  $X' = Y'$ . The main role of the encoder is to enhance the semantic relevance within a single modality. For example, for the text modality, self-attention enables the model to better capture the semantic connections between different utterances in the same text, such as the emotional echo relationship between preceding and following utterances in a dialogue, thereby making the semantic expression of unimodal features more complete. Formally,

$$X'_{X'} = \text{Transformer}_{\text{self}}(X', X', X') \in \mathbb{R}^{B \times N \times D} \quad (8)$$

**Cross-Transformer\_Info Mode (e.g., Text-Audio):** When the input modalities  $X' \neq Y'$ , the encoder will incorporate mutual information weights to prioritize the fusion of highly correlated cross-modal features. For instance, the description of “joy” in the text and the cheerful prosodic features in the audio have high mutual information, and cross-attention will focus on fusing these

features to improve the integration effect of cross-modal emotional information. Formally,

$$X'_{Y'} = \text{Transformer}_{\text{cross}}(X', Y', Y') \in \mathbb{R}^{B \times N \times D} \quad (9)$$

We further enhance the multi-head attention mechanism by integrating mutual information weights,

$$\text{Attn}(Q, K, V, MI) = \text{Softmax}\left(\frac{QK^T}{\sqrt{D_k}} + \log\left(\text{Softmax}\left(\frac{MI}{\tau}\right)\right)\right)V \quad (10)$$

where  $\tau$  denotes the temperature coefficient. Taking the text-audio modality pair as an example, if the semantics of “anger” in the text and the features such as angry tone and energy in the audio have high mutual information, then after processing, the mutual information  $MI$  will increase the attention score of this region, and the model will focus more on these highly correlated feature parts, thereby strengthening the capture of emotions.

Using modality selection weights  $[w_{ta}, w_{tv}, w_{av}]$ , we aggregate the cross-modal outputs:

$$\begin{aligned} t' &= t_t + w_{ta} \cdot a_t + w_{tv} \cdot v_t \\ a' &= a_a + w_{ta} \cdot t_a + w_{av} \cdot v_a \\ v' &= v_v + w_{tv} \cdot t_v + w_{av} \cdot a_v \end{aligned} \quad (11)$$

where  $t_t, a_a, v_v$  are Self-Transformer outputs, and  $a_t, v_t, t_a, v_a, t_v, a_v$  are Cross-Transformer\_Info outputs. Finally, the fusion module further filters and integrates the multimodal features processed through the above steps to remove redundant features that are unhelpful for emotional expression, and ultimately outputs the global fused feature  $u^y$ , which integrates highly correlated information from multiple modalities.

$$u^y = \sum_{m \in \{t, a, v\}} [\text{Softmax}(\text{MLP}_u(m')) \odot m'], m \in \{t, a, v\} \quad (12)$$

### 3.6 Speaker Personality Extractor

To better fuse and represent the semantic information between different modalities and capture the unique emotional expression patterns and styles of each speaker, we adopt a bidirectional LSTM to separately model and analyze the contextual information of the dialogue and the utterances of each speaker.

$$v_i^y, h_i = \overleftarrow{\text{LSTM}}(u_i^y, h_{i-1}); v_i^s, h_i' = \overrightarrow{\text{LSTM}}(u_i^s, h_{i-1}') \quad (13)$$

Here,  $v_i^y$  denotes the context-aware multimodal utterance embedding, while  $v_i^{sj}$  represents the utterance embedding specific to the  $j$ -th speaker's context. All  $v_i^{sj}$  are sequentially aggregated to form  $v_i^s$ . The final representation  $u_i'$  for the entire dialogue is obtained by concatenating  $v_i^y$  and  $v_i^s$ , integrating both global and speaker-specific cues.

### 3.7 Emotion Transfer Module

Existing multimodal dialogue emotion recognition models predominantly address single-utterance emotion classification, treating each utterance independently while overlooking the dynamic evolution of emotions throughout a dialogue. To capture these temporal dynamics, we introduce an Emotion Transfer Module based on Gated Recurrent Units (GRUs), which models emotional dependencies over time and outputs both an emotion transition probability matrix and the prediction of the next utterance's emotion.

**Emotion State Extraction:** We design an emotion state extractor that condenses the global fused feature  $u'$  into a compact emotional state vector, preserving essential emotional information while reducing computational overhead:

$$s_t = \text{MLP}_{\text{state}}(u') \in \mathbb{R}^{B \times N \times D_s} \quad (14)$$

where  $D_s$  is the dimension of the emotional state. This dimensionality reduction operation can effectively reduce the computational load in the subsequent temporal modeling process.

**Temporal Emotion Transition Modeling:** A GRU network is adopted to capture the temporal dependencies of emotional states, thereby predicting the emotion transition probability and the emotion of the next utterance:

$$h_t = \text{GRU}(s_t, h_{t-1}) \in \mathbb{R}^{B \times N \times D_h} \quad (15)$$

where  $D_h$  represents the GRU's hidden dimension. The next utterance's emotion is predicted by integrating the current emotional state  $s_t$  and the GRU output  $h_t$ :

$$y' = [h_t[:, :N-1, :] \oplus s_t[:, :N-1, :]] \quad (16)$$

$$P(s_{t+1}|s_t) = \text{Softmax}(\text{MLP}_{\text{trans}}(y')) \in \mathbb{R}^{B \times N \times 1 \times C}$$

**Transition Prediction Loss:** To optimize emotion transition modeling, we define a transition prediction loss based on negative log-likelihood:

$$\mathcal{L}_{\text{trans}} = -\frac{1}{N} \sum \log(P(s_{t+1}|s_t)[y'_{t+1}]) \quad (17)$$

This loss function calculates through negative log-likelihood, ensuring that the loss calculation is only performed on the emotion transition predictions of valid samples, thereby improving the accuracy of the model in predicting emotion transitions.

### 3.8 Classification

The final emotion prediction is obtained by applying the Softmax function to the integrated feature representation  $u'$ , with the highest probability category selected as the predicted emotion:

$$p = \text{Softmax}(u'), \quad y_i = \text{Argmax}_{k \in Y} (p_i[k]). \quad (18)$$

where  $p$  denotes the emotion probability distribution and  $y_i$  represents the predicted emotion category.

For optimization, we employ the negative log-likelihood loss function:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{B} \sum_{j=1}^B \sum_{k=1}^{B_j} y_{j,k} \log(\hat{y}_{j,k}) \quad (19)$$

where  $B$  indicates the total number of dialogue sessions,  $B_j$  represents the number of utterances in the  $j$ -th dialogue,  $y_{j,k}$  denotes the ground-truth emotion label, and  $\hat{y}_{j,k}$  refers to the predicted probability distribution for the  $k$ -th utterance in the  $j$ -th dialogue.

To balance the influence of different learning objectives, we introduce a multi-task loss function that combines classification, emotion transition prediction, and mutual information enhancement through carefully tuned weighting coefficients  $\gamma_1$ ,  $\gamma_2$ , and  $\gamma_3$ :

$$\mathcal{L} = \gamma_1 \mathcal{L}_{\text{cls}} + \gamma_2 \mathcal{L}_{\text{trans}} + \gamma_3 \mathcal{L}_{\text{MI}}. \quad (20)$$

This balanced optimization strategy enables the model to achieve strong performance in both utterance-level emotion classification and dialogue-level emotion transition prediction.

## 4 Experiment

### 4.1 Experimental Setup

**Datasets.** Experiments were conducted on two multimodal emotion recognition datasets: IEMOCAP [1] and MELD [19]. IEMOCAP contains approximately 12 hours of dyadic conversations from five pairs of actors, including both scripted and spontaneous dialogues, with emotion labels including Neutral, Happiness, Sadness, Anger, Frustrated, and Excited. MELD is constructed from dialogue clips of the TV series *Friends*, comprising 1,432 dialogues and 13,708 utterances, labeled with Neutral, Happiness, Surprise, Sadness, Anger, Disgust, and Fear. Table 1 summarizes the dataset statistics.

**Table 1: Statistics of the IEMOCAP and MELD Datasets**

| Dataset | Conversations |      |       | Utterances |       |        |
|---------|---------------|------|-------|------------|-------|--------|
|         | Train+Val     | Test | Total | Train+Val  | Test  | Total  |
| IEMOCAP | 120           | 31   | 151   | 5,810      | 1,623 | 7,433  |
| MELD    | 1,152         | 280  | 1,432 | 11,098     | 2,610 | 13,708 |

**Table 2: Comparison of Experimental Results. Data from the original paper are quoted directly in this paper. In each dataset, the best results are shown in bold, and dashes (“—”) are used to mark unreported results.**

| Models     |                                        | IEMOCAP      |              | MELD         |              |
|------------|----------------------------------------|--------------|--------------|--------------|--------------|
|            |                                        | ACC          | F1           | ACC          | F1           |
| Unimodal   | DialogueRNN (Majumder et al. 2019)[17] | 63.40        | 62.75        | —            | 57.03        |
|            | DialogueCRN (Hu et al. 2021)[7]        | 66.05        | 66.20        | 60.73        | 58.39        |
|            | DAG-ERC (Shen et al. 2021)[21]         | —            | 68.03        | —            | 63.65        |
|            | BLC (Xu et al. 2024)[27]               | —            | 68.22        | —            | 64.27        |
| Multimodal | JOYFUL (Li et al. 2023)[11]            | 70.55        | 71.03        | 62.53        | 61.77        |
|            | CMCF-SRNet (Zhang et al. 2023)[32]     | 70.50        | 69.60        | —            | 62.30        |
|            | SCMM (Yang et al. 2023)[28]            | —            | 67.53        | —            | 59.44        |
|            | MultiDAG+CL (Nguyen et al. 2024)[18]   | 69.11        | 69.08        | 64.41        | 64.00        |
|            | MAGTKD (Li et al. 2025)[12]            | 69.38        | 69.59        | 66.36        | 65.32        |
|            | PCDS (Shen et al. 2025)[20]            | 70.84        | 70.89        | 64.33        | 62.61        |
|            | ECERC (Zhang et al. 2025)[31]          | 71.60        | 71.78        | 67.32        | 66.46        |
| LLMs       | GPT-4o (Hurst et al. 2024)[9, 22]      | —            | 51.68        | —            | 59.18        |
|            | DeepSeek-R1 (Guo et al. 2025)[5, 22]   | —            | 53.64        | —            | 59.51        |
| Others     | MHGT+ECok (Wang et al. 2024)[25]       | —            | 71.50        | —            | 66.30        |
|            | ML-ERC (Kang et al. 2025)[10]          | —            | 66.09        | —            | 65.93        |
| Ours       | MIDE                                   | <b>74.86</b> | <b>74.80</b> | <b>67.74</b> | <b>66.76</b> |

**Baselines.** We compare the proposed method with several competitive baselines, which can be grouped into four categories:

**1. Unimodal Methods:** These approaches leverage only textual context and speaker information for emotion recognition, primarily based on recurrent neural networks (e.g., DialogueRNN [17], DialogueCRN [7]) or graph convolutional networks (e.g., DAG-ERC [21], BLC [27]).

**2. Multimodal Methods:** These models integrate text, audio, and visual modalities to enhance emotion understanding, including JOYFUL [11], CMCF-SRNet [32], SCMM [28], MultiDAG+CL [18], MAGTKD [12], PCDS [20], and ECERC [31].

**3. Large Language Models (LLMs):** We also include GPT-4o [9, 22], which supports cross-modal input and output for multilingual and audiovisual emotion tasks, and DeepSeek-R1 [5, 22], which employs a Mixture of Experts (MoE) architecture and reinforcement learning to improve emotional reasoning and domain-specific knowledge processing.

**4. Other Models:** This category includes MHGT+ECok [25], which augments emotional reasoning with a commonsense knowledge graph containing over 140K triples, and ML-ERC [10], a multi-label framework that employs pseudo multi-labels and MulWCL loss to address emotion shift and label confusion while preserving performance across single-label and multi-label scenarios.

**Evaluation Metrics.** We use Accuracy (ACC) and Weighted-F1 (F1) for evaluation, giving priority to Weighted-F1 as it more effectively reflects comprehensive performance across all emotion categories by accounting for class imbalance.

**Implementation Details.** To ensure reproducibility, we detail the preprocessing and training configurations. The input channel size of the 1D convolutional layer is adjusted according to the modality-specific feature dimensions of each dataset: for IEMOCAP, text = 1024, acoustic = 1582, visual = 342; for MELD, acoustic = 300, while text and visual dimensions remain consistent with IEMOCAP. The

model is trained with a batch size of 16, dropout rate of 0.5, and the Adam optimizer with a learning rate of  $1 \times 10^{-4}$ . All reported results are averaged over five independent runs.

## 4.2 Performance Analysis

The main experimental results are summarized in Table 2. Our proposed MIDE model achieves substantial improvements over all baseline categories on both IEMOCAP and MELD datasets, demonstrating its effectiveness in multimodal emotion recognition. Among unimodal approaches, text-based state-of-the-art methods like DAG-ERC and BLC show competitive performance, yet their accuracy and F1-scores remain considerably lower than MIDE. This performance gap highlights the inherent limitations of relying on a single modality to capture the complex affective dynamics in conversational data. Multimodal baselines, while integrating information from multiple sources, consistently fall short of MIDE’s performance. For instance, recent models like ECERC and MAGTKD show improved results over earlier multimodal approaches, but still underperform compared to our method. The superior performance of MIDE suggests that its dynamic modality selection and mutual information enhancement mechanisms more effectively leverage cross-modal interactions than fixed fusion strategies employed in these baselines. Notably, LLM-based approaches (GPT-4o and DeepSeek-R1) demonstrate relatively weak performance, indicating that general-purpose large language models struggle with fine-grained emotion recognition in dialogues without specialized architectural support. This contrasts sharply with MIDE’s strong results, which benefit from targeted components designed specifically for emotional dynamics. Evaluation against models in the “Other Models” category confirms MIDE’s favorable position. Overall, MIDE achieves the highest scores across all metrics, attaining 74.86% accuracy and 74.80% F1-score on IEMOCAP, and 67.74% accuracy and 66.76% F1-score on MELD. These results validate the effectiveness of our integrated framework—particularly its dynamic modality selection,

**Table 3: Display Performance Metrics on IEMOCAP and MELD. (We report precision, recall, f1-score (per class), accuracy, macro average, and weighted average.)**

| (a) IEMOCAP  |           |        |               |         |
|--------------|-----------|--------|---------------|---------|
|              | precision | recall | f1-score      | support |
| Happy        | 0.6403    | 0.6181 | 0.6290        | 144     |
| Sad          | 0.8208    | 0.8041 | 0.8124        | 245     |
| Neutral      | 0.7410    | 0.8047 | 0.7715        | 384     |
| Angry        | 0.7261    | 0.6706 | 0.6972        | 170     |
| Excited      | 0.8188    | 0.8161 | 0.8174        | 299     |
| Frustrated   | 0.7043    | 0.6877 | 0.6959        | 381     |
| accuracy     | —         | —      | <b>0.7486</b> | 1623    |
| macro avg    | 0.7419    | 0.7335 | 0.7372        | 1623    |
| weighted avg | 0.7483    | 0.7486 | <b>0.7480</b> | 1623    |

| (b) MELD     |           |        |               |         |
|--------------|-----------|--------|---------------|---------|
|              | precision | recall | f1-score      | support |
| Neutral      | 0.7710    | 0.8392 | 0.8037        | 1256    |
| Surprise     | 0.5978    | 0.5765 | 0.5870        | 281     |
| Fear         | 0.2759    | 0.1600 | 0.2025        | 50      |
| Sadness      | 0.6087    | 0.3365 | 0.4334        | 208     |
| Joy          | 0.6613    | 0.6169 | 0.6384        | 402     |
| Disgust      | 0.4000    | 0.2353 | 0.2963        | 68      |
| Anger        | 0.5085    | 0.6087 | 0.5541        | 345     |
| accuracy     | —         | —      | <b>0.6774</b> | 2610    |
| macro avg    | 0.5462    | 0.4819 | 0.5022        | 2610    |
| weighted avg | 0.6687    | 0.6774 | <b>0.6676</b> | 2610    |

**Table 4: Results of Ablation Experiment**

| Models     | IEMOCAP      |              | MELD         |              |
|------------|--------------|--------------|--------------|--------------|
|            | ACC          | F1           | ACC          | F1           |
| - w/o DMS  | 69.69        | 70.32        | 66.05        | 65.14        |
| - w/o MIE  | 70.73        | 71.25        | 66.59        | 65.26        |
| - w/o ETM  | 73.94        | 74.07        | 67.01        | 66.18        |
| - w/o SPE  | 71.97        | 72.23        | 66.17        | 65.62        |
| Full Model | <b>74.86</b> | <b>74.80</b> | <b>67.74</b> | <b>66.76</b> |

mutual information enhancement, and explicit modeling of speaker characteristics and emotion transitions—in addressing the complex challenges of multimodal dialogue emotion.

Furthermore, as shown in Table 3, our method offers a fine-grained breakdown of precision and F1-score per emotion category on both IEMOCAP and MELD datasets. This category-level analysis provides clearer visualization of our approach’s strengths across diverse emotion types and data distributions, while enabling direct comparison with other methods at the emotion-specific level to support targeted model optimization.

### 4.3 Ablation Study

To evaluate the contribution of each core component in our method, we conduct a series of ablation experiments. By progressively removing key modules and observing the corresponding performance

**Table 5: Results of Fixed Modality Comparative Experiments**

| Models                | IEMOCAP      |              | MELD         |              |
|-----------------------|--------------|--------------|--------------|--------------|
|                       | ACC          | F1           | ACC          | F1           |
| Text                  | 65.37        | 65.61        | 66.21        | 64.78        |
| Audio                 | 58.78        | 58.88        | 47.93        | 35.32        |
| Visual                | 36.35        | 35.42        | 48.12        | 31.27        |
| Text + Audio          | 71.16        | 71.58        | 66.74        | 65.50        |
| Text + Visual         | 70.18        | 70.69        | 66.51        | 65.34        |
| Audio + Visual        | 62.97        | 62.54        | 47.93        | 37.49        |
| Text + Audio + Visual | 70.73        | 71.25        | 66.59        | 65.26        |
| Ours                  | <b>74.86</b> | <b>74.80</b> | <b>67.74</b> | <b>66.76</b> |

changes, we quantitatively assess the role of each proposed mechanism. The results are summarized in Table 4, from which we can observe:

1) Removing **Dynamic Modality Selection (DMS)** causes noticeable degradation, particularly on IEMOCAP. This module adaptively evaluates inter-modal correlations via mutual information and assigns dynamic fusion weights, effectively suppressing noisy modalities. In contrast, fixed fusion strategies lack adaptability to modality quality variations.

2) Ablating **Mutual Information Enhancement (MIE)** significantly reduces F1-score on both datasets, especially on IEMOCAP. The module integrates mutual information maximization across modality selection, attention allocation, and loss optimization, enhancing semantic alignment while reducing modality heterogeneity through adaptive temperature adjustment.

3) Removing **Emotion Transfer Module (ETM)** hurts performance. By modeling temporal dependencies via GRUs and emotion transition probabilities, it enables coherent dialogue-level recognition—addressing a key limitation of utterance-isolated approaches.

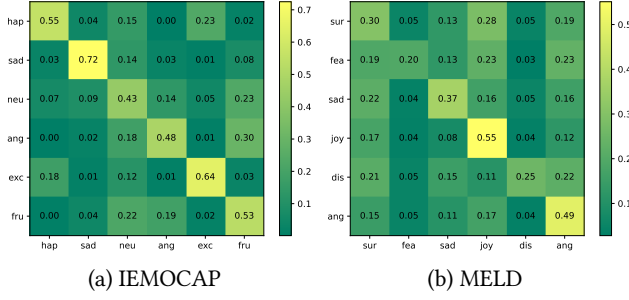
4) Without **Speaker Personality Extractor (SPE)**, performance consistently declines. The module uses bidirectional LSTMs to encode speaker-specific patterns, improving consistency in multi-speaker scenarios with minimal overhead, whereas conventional methods overlook such individuality.

Overall, the ablation study confirms that each proposed module contributes distinctively to the model’s performance. The synergy among these components enables robust, accurate, and context-aware emotion recognition.

### 4.4 Fixed Modality Comparative Experiments

To examine the assumption that “more modalities lead to better performance” in multimodal dialogue emotion recognition, and to validate the effectiveness of our dynamic modality selection mechanism, we conduct comparative experiments with fixed modality combinations. We systematically compare unimodal, bimodal, and trimodal settings using equal-weight fusion to assess the actual contribution of each modality combination.

Results in Table 5 show that in unimodal settings, text modality significantly outperforms audio and visual modalities on both datasets, confirming that emotional understanding in dialogues relies primarily on textual semantics, while audio and visual cues are more susceptible to noise and offer limited discriminative power



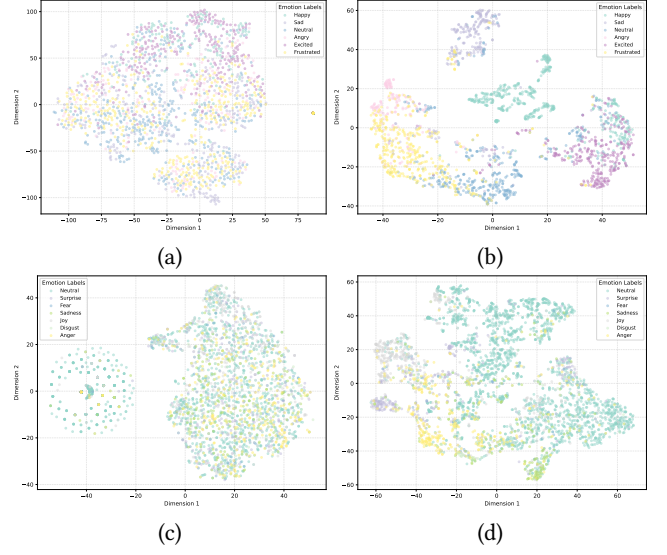
**Figure 2: Emotional transfer probability heatmap. (The vertical axis represents the emotion of the current utterance, while the horizontal axis represents the emotion of subsequent utterances.)**

alone. Among bimodal combinations, Text+Audio achieves the best performance, indicating strong complementarity between textual content and acoustic features. In contrast, Audio+Visual performs poorly, suggesting high redundancy and weak complementarity between these two modalities. Notably, the trimodal combination (Text+Audio+Visual) fails to surpass the best bimodal setup (Text+Audio), demonstrating that incorporating all modalities can introduce noise and redundancy, thereby challenging the assumption that “more modalities are always better”.

Our method substantially outperforms all fixed fusion strategies across both datasets. This advantage stems from the dynamic modality selection mechanism, which adaptively adjusts fusion weights based on modality quality and complementarity. By effectively leveraging complementary information while suppressing noisy or redundant modalities, our approach achieves a significant performance improvement.

#### 4.5 Emotional Transfer Matrix Visualization

Figure 2 presents the emotion transition probability heatmaps for the IEMOCAP and MELD datasets, demonstrating that our proposed emotion transition module effectively captures dynamic emotion patterns in conversations. The visualization reveals several key findings: 1) In the IEMOCAP heatmap, the module accurately identifies strongly correlated emotion transitions that align with real-world emotional logic. For example, when the current emotion is excitement (exc), the probability of remaining excited reaches 64%, significantly higher than the 1% transition probability to sadness (sad). The module also effectively suppresses improbable transitions, such as the zero probability of happiness (hap) transitioning to anger (ang), demonstrating its capability to filter out spurious temporal dependencies. 2) The MELD heatmap shows that the module maintains robust performance even with different emotion categories, including surprise (sur) and fear (fea). This indicates strong generalization ability across diverse emotional contexts and dataset characteristics. 3) By leveraging GRU-based temporal modeling, our module not only captures semantically reasonable emotion transitions but also effectively suppresses illogical emotional jumps. This approach addresses a key limitation of traditional methods that ignore temporal emotional dependencies, while enabling dynamic emotion prediction based on static classification results. The



**Figure 3: (a) and (c) show the initial utterance vectors for the IEMOCAP and MELD datasets, respectively; (b) and (d) display the vectors processed by our model.**

demonstrated capability provides crucial support for coherent emotion recognition in extended dialogues, validating the effectiveness of our emotion transition modeling strategy.

#### 4.6 Result Visualization

Figure 3 demonstrates that our model effectively reorganizes utterance representations in the embedding space. Before processing, emotion vectors (labeled as “sad”, “neu”, “hap”, etc.) appear randomly scattered without clear clustering patterns. After transformation by our model, vectors of the same emotion category form compact, well-separated clusters. This structural enhancement indicates that our approach successfully captures discriminative emotional features and maps semantically similar utterances to proximal regions in the latent space. The observed clustering quality provides visual evidence of the model’s superior capability in emotion feature extraction and representation learning, which directly contributes to its recognition performance.

### 5 Conclusion

This paper addresses key limitations in multimodal dialogue emotion recognition, including rigid fusion strategies, cross-modal semantic gaps, and inadequate modeling of temporal dependencies. We propose MIDE, a novel framework that integrates mutual information enhancement with dynamic modality selection to enable adaptive, quality-aware fusion of multimodal signals. Through mutual information-guided attention mechanisms and dedicated emotion transition modeling with GRUs, our approach achieves deeper semantic alignment and captures contextual emotional dynamics effectively. Comprehensive evaluations on IEMOCAP and MELD datasets demonstrate that MIDE consistently outperforms state-of-the-art methods across all metrics, validating its effectiveness in handling complex dialogue scenarios.

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